

Mechanical Designer involved with many successful consumer products. Key strengths:

- Conceive and execute innovative solutions in alignment with design intent, business objectives, regulatory requirements, quality / safety standards, manufacturing logistics, etc.
- Develop processes to increase performance, efficiency, control costs, and reduce defects
- Building and maintaining supplier relations
- Navigate complex (often conflicting) constraints to preserve Industrial Design integrity, while ensuring manufacturability, with an eye on efficiency and yield
- Strong communication and negotiation skills
- Extensive part design experience
- Leverage my efforts for the team win

Experience across a wide range of product development: Disc Changers to Salad Robots.

PROFESSIONAL EXPERIENCE

MECHANISTIC – Redwood City, California **Senior Mechanical Designer** 08/22 – present

Product development and engineering consultant.

Provide engineering and design-related support services for a wide range of clients.

HYPHEN – San Jose, California **Staff Mechanical Engineer** 05/21 – 08/22

Designed, validated, and produced multiple hardware solutions and systems for Hyphen's automated makeline. Contributions included:

- design and development of portion cup dispensing subsystem
- concept through production of 'backsplash' electronics enclosure
- detailed design and production of multiple Gen 1 1/3-pan ingredient dispensers
- detailed design and production of Gen 1 1/1-pan ingredient dispensers
- detailed design of modular dispensing system to maximize configuration flexibility, minimize part count
- lead Gen 2 ME/EE development - coordinated program-level efforts, facilitated collaboration, enforced aggressive schedule, etc.
- manage and liaise with 50+ external contract resources
- Gen 2 ingredient dispenser concept validation testing - 1+ new concepts in-test per week

MECHANISTIC – Redwood City, California **Senior Mechanical Designer** 05/18 – 05/21

Sole proprietor of design consultancy providing product development services from concept through manufacturing support.

Provide design-related support services for a wide range of clients. ***Selected projects include:***

- IOT module for Bay Wheels ebike.
- Engineering consultation and design support for connected Pet scale.
- System architecture through detailed part design for ecargo bike.
- Concept through production design and engineering support for high performance LED lighting system.

Erik Norris – Staff Mechanical Engineer

NOON HOME – Cupertino, California

Senior Mechanical Engineer

03/17 – 05/18

Worked within a small, high-caliber team to launch a connected light switch. Supported first-gen products through ramp into channel. Completed design validation and release to-tooling of second-gen product. Selected contributions:

- Intensive DFM with strong focus on cost/quality
- Stringent design validation, Reliability and Environmental test program
- Wide-ranging design explorations in support of roadmap definition
- Fault isolation, component-level troubleshooting, NPS improvement
- Define process and communicate intent to external partners - CMF, workmanship, assembly details, etc.
- Draft and enforce specifications, procedures and process into diverse supply chain
- Collaborate with internal stakeholders and external resources to find the most efficient means. repeat.

AVEGANT – Belmont, California

Senior Mechanical Engineer

06/15 – 03/17

Contributed to the development of cutting-edge near eye display technology. Responsible for mechanical design and development of video headset.

Worked closely with hardware and industrial design teams. Product development from ideation to product launch. ***Selected contributions:***

- Design and development of mechanical components and mechanisms
- Structural, kinematics, and tolerance analysis of parts and systems
- 3D design and 2D drafting of injection molded, machined, stamped, and wire-form components
- Tolerance analysis of complex assemblies and precision component parts
- Interface with component part manufacturing partners on design manufacturability and resolution of mechanical issues
- Author clear documentation on mechanical inspection and test requirements
- Collaborate with Industrial Designers to develop human interface components
- Communicate with overseas manufacturers and suppliers to address manufacturing related issues
- Provide engineering direction to supporting engineers
- Travel to overseas manufacturing facilities as required

KALEIDESCAPE – Sunnyvale, California

Senior Mechanical Design Engineer

05/08 – 03/15

Sole mechanical design engineer responsible for all aspects of product design - NPI to sustaining.

Manage all aspects of mechanical design for new products, providing design-related direction across the entire product development and enhancement lifecycle. Hold Industrial Design responsibilities, including concept generation, sketches, renderings, prototyping, and design implementation. Identify, diagnose, and resolve root cause of emerging production issues. Maintain manufacturing partner relations to ensure product development in alignment with established standards. ***Selected contributions:***

- Define, design, model, prototype, test, refine, document, release, sustain. Rinse, repeat.
- Create and maintain mechanical designs and specifications for entire product family.
- Evaluate existing products for cost-reduction or manufacturing process improvement opportunities.
- Apply sustaining engineering practices with a focus on eliminating field failures.
- Products (shipped in bold): **1U Server, Speed Reader, 3U Server, Mini System, M500 Player, M300 Player, Modular Disc Vault, M700 Disc Changer, Cinema One.**

Erik Norris – Staff Mechanical Engineer

DIGIDESIGN – Daly City, California

Senior Mechanical Designer

2003 – 2008

Led a wide array of product design and engineering efforts for digital audio workstations.

Directed multiple, concurrent product design efforts, assuming project management-level responsibility and performing ongoing implementation of process improvements. Served as system administrator and subject matter expert for Pro/ENGINEER and Pro/INTRALINK (Windchill). ***Selected contributions:***

- Created improved documentation and design workflow across all departmental operations.
- Configured Windchill solutions for CAD collaboration and review by internal non-CAD users.
- Implemented revision control and change management procedures for CAD deliverables, utilizing Pro/PDM, Pro/INTRALINK (now Windchill) to manage product data.
- Developed and maintained internal engineering department website for collaboration and exchange.
- Products (shipped in bold): **002, Mbox, 002 Rack, ICON, D-Command, D-Show, Mbox 2, Mbox 2 Pro, Mbox 2 Mini, 003, 003 Rack, Mbox 2 Micro, C|24.**

MECHANISTIC – San Jose, California

Senior Mechanical Designer

1999 – 2003

Sole proprietor of design consultancy providing product development services from concept through manufacturing support.

Directed design, development, and launch efforts spanning the entire product lifecycle: brainstorming, testing, quality assurance, and manufacturing. Integrated with distributed team members, interfacing with multiple client groups, suppliers, etc. Collaborated throughout the organization to coordinate and execute design, engineering, and process efforts. Ensured consistent adherence to project, budget and schedule objectives.

Selected contributions:

- Won high-profile clients through consistent demonstration of value proposition and cradle-to-grave product development expertise. Clients included Microsoft, Motorola, HP, SGI, D2M, FITCH, WebTV.
- Provided freelance services in all phases of product development, from concept to production.
- Coordinated detailed design and documentation with successful transfer to high-volume manufacturing.
- Collaborated with suppliers and manufacturers to ensure design integrity, quality control, yield, etc.
- Products (some withheld, shipped in bold): **Microsoft Xbox, Microsoft Ultimate TV Media Center, Motorola MP3 Player, Netscreen 500, HP Jornada 430 PDA, SGI 540 Visual Workstation, DVDO Portable DVD Player, Echelon LonWorks network controllers, WebTV Media Center / Receivers.**

FROGDESIGN – Sunnyvale, California

Mechanical Designer

1997 – 1999

Delivered innovative design solutions for a wide variety of clients, across all phases of product development: from brainstorming through manufacturing.

Evaluated early-stage concepts with feasibility studies, prototyping, and testing. Refined designs in CAD, preserving design intent while ensuring function. Balanced numerous (often conflicting) constraints in pursuit of the best, most appropriate design. ***Selected contributions:***

- Worked closely with industrial designers to maintain subtle surface, gap and material finish requirements.
- Captured the design intent in CAD and ensured it was realized in the final manufactured product.
- Supported plastic injection molding and sheet metal tooling design, pilot builds, and manufacturing ramp.
- Products (some withheld, shipped in bold): **KaVo X-Unit / Edeca Dental System, Newscorp Satellite TV Receiver, Vadem Clio.**